Note: number of ν_e appearance events in long-baseline experiments

1 General expression

The number of ν_β charged-current (CC) events produced by a ν_α beam and selected in a far detector at a distance L from the source is (see, e.g., Ref. [1])

$$N_{\alpha\to\beta} = \mathrm{POT}\; N_T \int dE\; \frac{d\Phi_\alpha}{dE}(E)\; P_{\alpha\to\beta}(E,L)\; \sigma_\beta^{CC}(E)\; \varepsilon_\beta(E), \tag{1}$$

where

- $d\Phi_\alpha/dE$ is the unoscillated ν_α flux at the far detector, per proton on target and per unit energy (units: $\nu\ \mathrm{cm}^{-2}\ \mathrm{GeV}^{-1}\ \mathrm{POT}^{-1}$);
- POT is the number of protons on target delivered in the beam mode considered (the exposure);
- N_T is the number of target nucleons in the fiducial volume;
- $P_{\alpha\to\beta}$ is the oscillation probability in matter;
- σ_β^{CC} is the CC cross section per nucleon of ν_β, in cm^2;
- ε_β is the selection efficiency.

In a full analysis Eq. (1) is also convolved with the detector energy response and backgrounds are added; neither is included here.

Number of targets. For a detector of fiducial mass M, counting one nucleon per atomic mass unit,

$$N_T = \frac{M}{m_u} = M\,[\mathrm{g}]\; N_A \;\simeq\; 6.022\times10^{32}\times\frac{M}{1\,\mathrm{kt}}, \tag{2}$$

with $1\,\mathrm{kt} = 10^9\,\mathrm{g}$ and $N_A = 6.022\,140\,76\times10^{23}\,\mathrm{mol}^{-1}$ [3].

Two independent factors: beam and detector. POT and N_T are not related to each other: the first describes the beam, the second the detector. Eq. (1) factorizes as

$$N = \underbrace{\left(\mathrm{POT}\times\frac{d\Phi}{dE}\right)}_{\text{beam: fluence } [\nu\,\mathrm{cm}^{-2}\,\mathrm{GeV}^{-1}]} \times \underbrace{\left(N_T\times\sigma^{CC}\right)}_{\text{detector: target area } [\mathrm{cm}^2]} \times P\times\varepsilon. \tag{3}$$

The flux per POT, $d\Phi/dE$, is a property of the beamline (horns, target, decay pipe) and of the distance ($\propto 1/L^2$) and off-axis angle of the detector; multiplied by the POT it gives the *fluence*, the number of neutrinos crossing each cm^2 of the detector. $N_T\,\sigma^{CC}$ is the total cross-sectional area that the target nucleons offer to a neutrino. Their product is the expected number of interactions.

POT, beam power and running time. A beam of protons of energy E_p delivered with average power P_beam for an effective running time t accumulates

$$\mathrm{POT} = \frac{P_\mathrm{beam}\,t}{E_p\,[\mathrm{J}]} = 6.24\times10^{20}\times\frac{P_\mathrm{beam}}{1\,\mathrm{MW}}\times\frac{t}{10^7\,\mathrm{s}}\times\frac{100\,\mathrm{GeV}}{E_p}, \tag{4}$$

using $1\,\mathrm{GeV} = 1.602\times10^{-10}\,\mathrm{J}$ [3]. Examples:

- Hyper-K: $E_p = 30\,\mathrm{GeV}$, $P_\mathrm{beam} = 1.3\,\mathrm{MW}$, $10\times10^7\,\mathrm{s}$ give 2.7×10^{22} POT, the Design Report exposure [8];

- DUNE: $E_p = 120\,\text{GeV}$, $P_{\text{beam}} = 1.2\,\text{MW}$ and $1.1 \times 10^{21}\,\text{POT}$ per year [7], which by Eq. (4) corresponds to an effective $1.8 \times 10^7\,\text{s}$ of beam per calendar year.

The exposure of a long-baseline experiment is therefore often quoted as mass × power × time, e.g. in kt·MW·year (624 kt·MW·year for DUNE [7]), since $N \propto M \times \text{POT} \propto M\,P_{\text{beam}}\,t$.

2 Simplified one-energy formula

2.1 Assumptions

1. **Pure beam**: the neutrino (antineutrino) beam contains only ν_μ ($\bar{\nu}_\mu$). The wrong-sign component and the intrinsic ν_e contamination of the beam are neglected, and only the right-sign appearance signal is counted.
2. **One energy**: all neutrinos have the energy E_0, defined here as the flux-weighted mean energy of the main peak of the beam (energy bins with a flux above 10 % of its maximum), computed from the public fluxes of Sec. 3.
3. **Total flux**: $\Phi = \int (d\Phi/dE)\,dE$ is the total ν_μ ($\bar{\nu}_\mu$) flux at the far detector.
4. **One cross section**: the inclusive CC cross section per nucleon at E_0.
5. **Effective efficiency**: a single number ε_{eff} per experiment and beam mode, fixed on published predictions (Sec. 4).
6. No backgrounds, no energy resolution, constant matter density.

2.2 The formula

With these assumptions Eq. (1) becomes

$$\boxed{N = \varepsilon_{\text{eff}}\; P(\nu_\mu \to \nu_e; E_0, L)\; \sigma^{CC}(E_0)\; \Phi\; \text{POT}\; N_T} \tag{5}$$

Dimensional check:

$$[\text{cm}^2] \times [\nu\,\text{cm}^{-2}\,\text{POT}^{-1}] \times [\text{POT}] \times [\text{nucleons}] = \text{events}. \tag{6}$$

Expressing each factor in its natural unit gives a practical version,

$$N \simeq 60.2\;\varepsilon_{\text{eff}}\,P \left[\frac{\sigma^{CC}}{10^{-38}\,\text{cm}^2}\right] \left[\frac{\Phi}{10^{-14}\,\text{cm}^{-2}\,\text{POT}^{-1}}\right] \left[\frac{\text{POT}}{10^{21}}\right] \left[\frac{M}{\text{kt}}\right], \tag{7}$$

since $10^{-38} \cdot 10^{-14} \cdot 10^{21} \cdot 6.022 \times 10^{32} = 60.2$. With $P = 1$ and $\varepsilon_{\text{eff}} = 1$, Eq. (5) gives the number of CC interactions, $N_{P=1}$, there would be if every ν_μ became a ν_e.

3 Ingredients

Fluxes. T2K: 2020 public flux release at Super-Kamiokande [4, 5], also used for Hyper-Kamiokande, which shares the J-PARC beam, off-axis angle and baseline [8]. NOvA: 2017 public near-detector flux [6], scaled to the far detector as $(L_{\text{ND}}/L_{\text{FD}})^2 = (1\,\text{km}/810\,\text{km})^2$ (no far-detector flux is public). DUNE: far-detector flux of the TDR configuration [7].

Cross sections. For T2K, Hyper-K and NOvA, the standard inclusive CC cross section distributed with GLoBES [1, 2] (file `XCC.dat`), which is taken at low energy from the Super-Kamiokande cross sections for water [9] and at high energy from Ref. [10]. For DUNE, the GENIE [11] cross section on argon of the TDR configuration [7]. At high energy $\sigma^{CC}/E \to 0.67$ (ν) and 0.34 ($\bar{\nu}$) $\times 10^{-38}\,\text{cm}^2\,\text{GeV}^{-1}$ [12].

Probabilities. Three-flavor oscillations in constant-density matter computed with the NuFast algorithm [13].

Detectors and exposures. Baselines, densities, fiducial masses and POT from the references in Table 1.

Table 1: Ingredients of Eq. (5). E_0, Φ and $\sigma^{CC}(E_0)$ are computed in this note from the fluxes and cross sections above. References: T2K [14, 15], NOvA [16, 17], Hyper-K [8], DUNE [7].

	beam	L [km]	ρ [g/cm^3]	M [kt]	POT	E_0 [GeV]	Φ [cm^{-2} POT^{-1}]	$\sigma^{CC}(E_0)$ [cm^2]	N_T
T2K	ν	295	2.6	22.5	1.494×10^{21}	0.57	1.207×10^{-14}	4.07×10^{-39}	1.35×10^{34}
	$\bar{\nu}$				1.635×10^{21}	0.55	9.76×10^{-15}	1.16×10^{-39}	
Hyper-K	ν	295	2.6	187	6.75×10^{21}	0.57	1.207×10^{-14}	4.07×10^{-39}	1.13×10^{35}
	$\bar{\nu}$				2.025×10^{22}	0.55	9.76×10^{-15}	1.16×10^{-39}	
NOvA	ν	810	2.84	14	2.661×10^{21}	1.80	1.806×10^{-14}	1.43×10^{-38}	8.43×10^{33}
	$\bar{\nu}$				1.25×10^{21}	1.79	1.467×10^{-14}	6.31×10^{-39}	
DUNE	ν	1300	2.848	40	7.15×10^{21}	2.31	1.548×10^{-14}	1.78×10^{-38}	2.41×10^{34}
	$\bar{\nu}$				7.15×10^{21}	2.27	1.385×10^{-14}	7.29×10^{-39}	

4 Effective efficiency

ε_{eff} is fixed by requiring that Eq. (5), evaluated with the oscillation parameters and exposure of a published prediction, reproduces the published number of signal events N_{ref}:

$$\varepsilon_{\text{eff}} = \frac{N_{\text{ref}}}{N_{\varepsilon=1}(\text{reference parameters})}. \tag{8}$$

The reference predictions are listed in Table 2:

- **Hyper-K**: expected $\nu_\mu \to \nu_e$ (ν beam) and $\bar{\nu}_\mu \to \bar{\nu}_e$ ($\bar{\nu}$ beam) signal events, Tables XXXVI–XXXVII of the Design Report [8], for $\sin^2 2\theta_{13} = 0.1$, $\delta_{CP} = 0$, $\sin^2\theta_{23} = 0.5$, $\Delta m^2_{32} = 2.4 \times 10^{-3}\,\text{eV}^2$, $\sin^2 2\theta_{12} = 0.8704$, $\Delta m^2_{21} = 7.6 \times 10^{-5}\,\text{eV}^2$, normal ordering.

- **NOvA**: right-sign appearance signal at the NOvA best fit, Table 2 of Ref. [17]: normal ordering, $\delta_{CP} = 0.87\pi$, $\sin^2\theta_{23} = 0.547$, $\Delta m^2_{32} = 2.441 \times 10^{-3}\,\text{eV}^2$, $\sin^2 2\theta_{13} = 0.0851$ [18]; solar parameters $\sin^2 2\theta_{12} = 0.851$, $\Delta m^2_{21} = 7.53 \times 10^{-5}\,\text{eV}^2$ [19].

- **T2K**: Table XI of Ref. [15] gives the total number of single-ring e-like events (signal + background) for four values of δ_{CP}, with the other parameters fixed to Table III of the same reference ($\sin^2\theta_{23} = 0.528$, $\sin^2\theta_{13} = 0.0212$, $\sin^2\theta_{12} = 0.304$, $\Delta m^2_{32} = 2.509 \times 10^{-3}\,\text{eV}^2$, $\Delta m^2_{21} = 7.53 \times 10^{-5}\,\text{eV}^2$, normal ordering). Since the background does not depend on δ_{CP} to first order, the four values are fitted with $N_{\text{tot}}(\delta_{CP}) = \varepsilon_{\text{eff}}\, N_{\varepsilon=1}(\delta_{CP}) + B$, which gives ε_{eff} and the background B ($B = 19.6$ and 12.3 events in the ν and $\bar{\nu}$ beams). In the $\bar{\nu}$ beam the fit also absorbs the wrong-sign $\nu_\mu \to \nu_e$ contribution, which has the opposite δ_{CP} dependence, so ε_{eff} is less reliable there.

- **DUNE**: no table of event numbers is published with the TDR configuration; N_{ref} is the right-sign appearance signal computed with the official GLoBES configuration of Ref. [7] (flux, GENIE cross sections, detector response; 6.5 + 6.5 years, 40 kt), for the parameters used in that reference: NuFIT 4.0 [20] normal ordering and $\delta_{CP} = 0$.

Table 2: Calibration of the effective efficiency on published predictions.

	beam	$N_{\varepsilon=1}$	N_{ref}	ε_{eff}	reference
T2K	ν	fit (4 values)	74.4 / 62.2 / 50.6 / 62.7	0.89	[15], Table XI
	$\bar{\nu}$	fit (4 values)	17.1 / 19.6 / 21.7 / 19.3	0.72	[15], Table XI
Hyper-K	ν	2063.8	1643	0.80	[8], Tables XXXVI–XXXVII
	$\bar{\nu}$	1158.0	1183	1.02	[8], Tables XXXVI–XXXVII
NOvA	ν	291.7	125.3	0.43	[17], Table 2
	$\bar{\nu}$	37.1	18.9	0.51	[17], Table 2
DUNE	ν	3493.6	2180.3	0.62	[7], GLoBES configuration
	$\bar{\nu}$	407.6	437.6	1.07	[7], GLoBES configuration

Interpretation. ε_{eff} is a normalization, not a measured detector efficiency. It can be factorized as

$$\varepsilon_{\text{eff}} = \underbrace{\frac{N_{\text{ref}}}{\text{POT}\, N_T \int \Phi(E)\, \sigma(E)\, P(E)\, dE}}_{\varepsilon_{\text{det}}} \times \underbrace{\frac{\int \Phi(E)\, \sigma(E)\, P(E)\, dE}{\Phi\, \sigma(E_0)\, P(E_0)}}_{c_{\text{mono}}}, \tag{9}$$

i.e., the selection efficiency ε_{det} times a correction c_{mono} for the one-energy approximation; ε_{det} also absorbs any difference between the cross sections used here and those of the collaborations. Table 3 gives c_{mono}, computed with the flux histograms, the cross sections and the probabilities of Sec. 3 at the reference parameters, and the resulting ε_{det}.

Table 3: Factorization of ε_{eff} into the one-energy correction c_{mono} and $\varepsilon_{\text{det}} = \varepsilon_{\text{eff}}/c_{\text{mono}}$ (reference parameters of Table 2; T2K at $\delta_{CP} = -\pi/2$).

	ν beam			$\bar{\nu}$ beam		
	ε_{eff}	c_{mono}	ε_{det}	ε_{eff}	c_{mono}	ε_{det}
T2K	0.89	0.81	1.10	0.72	1.06	0.68
Hyper-K	0.80	0.85	0.94	1.02	1.00	1.02
NOvA	0.43	0.79	0.54	0.51	0.83	0.61
DUNE	0.62	0.78	0.79	1.07	1.39	0.77

In the ν beams E_0 lies close to the oscillation maximum, while the flux is spread over energies where P is smaller, so $c_{\text{mono}} \approx 0.8$ and ε_{eff} is lower than the detector efficiency. In the $\bar{\nu}$ beams the antineutrino cross section grows quickly with energy, so the high-energy part of the flux weighs more: this compensates the lower P at T2K and Hyper-K ($c_{\text{mono}} \approx 1$) and dominates at DUNE ($c_{\text{mono}} \approx 1.4$). This, and not the beam contamination (the Hyper-K, NOvA and DUNE references are right-sign signal only), is why ε_{eff} can exceed 1. The values of ε_{det} above 1 (T2K ν, Hyper-K $\bar{\nu}$) show that ε_{det} is not a pure selection efficiency either: it also contains the differences in cross sections and in the definition of the samples.

Using $\varepsilon_{\text{eff}} = 1$. Setting $\varepsilon_{\text{eff}} = 1$ gives the number of ν_e ($\bar{\nu}_e$) CC interactions in the fiducial volume for a perfect detector and a monochromatic beam. This needs no calibration and is the choice of the tutorial `tutorial_event_counting.ipynb`; the published predictions are then used as a check, with ε_{eff} of Table 2 playing the role of the ratio $N_{\text{ref}}/N_{\varepsilon=1}$. The bi-probability plots and the shape of the δ_{CP} dependence do not depend on this choice. The numbers of events grow by $1/\varepsilon_{\text{eff}}$ (a factor of about 2 for NOvA) and the significances become optimistic: with statistical errors only, the maximum sensitivity to CP violation (true NO) is 2.5σ (T2K), 4.4σ (NOvA), 17σ (Hyper-K) and 18σ (DUNE), instead of 2.3, 2.9, 16 and 16σ with the calibrated values; with a 5 % normalization uncertainty the Hyper-K and DUNE values (8.7σ and 9.3σ) are almost unchanged.

5 Worked example: T2K, neutrino beam

With Table 1 and Eq. (7), at the NuFIT 6.0 normal-ordering best fit [21],

$$N_{P=1} = 60.2 \times 0.407 \times 1.207 \times 1.494 \times 22.5 \simeq 994, \tag{10}$$

$$N_{\varepsilon=1} = P(E_0)\, N_{P=1} = 0.0533 \times 994 \simeq 53, \tag{11}$$

$$N = \varepsilon_{\text{eff}}\, N_{\varepsilon=1} = 0.89 \times 53 \simeq 47. \tag{12}$$

Table 4: Number of right-sign appearance signal events from Eq. (5) with the effective efficiencies of Table 2, at the NuFIT 6.0 normal-ordering best fit [21] and the exposures of Table 1.

	beam	$N_{P=1}$	$P(E_0)$	$N_{\varepsilon=1}$	ε_{eff}	N
T2K	ν	994	0.0533	53.0	0.89	47.2
	$\bar{\nu}$	250	0.0316	7.9	0.72	5.7
Hyper-K	ν	37336	0.0533	1990.5	0.80	1592.4
	$\bar{\nu}$	25773	0.0316	813.7	1.02	830.0
NOvA	ν	5811	0.0547	317.7	0.43	136.6
	$\bar{\nu}$	975	0.0249	24.3	0.51	12.4
DUNE	ν	47547	0.0738	3509.6	0.62	2175.9
	$\bar{\nu}$	17380	0.0178	310.1	1.07	331.8

6 Sensitivity to CP violation

The question is: if the true value is δ_{CP}, how strongly can an experiment that measures N_ν and $N_{\bar{\nu}}$ exclude CP conservation? The test statistic is built in five steps.

Step 1: likelihood of a counting experiment. The number of events D_i observed in sample i ($i = \nu, \bar{\nu}$) is Poisson distributed around the prediction T_i of a hypothesis. The two beam modes are independent, so the likelihood is the product

$$\mathcal{L}(T) = \prod_{i=\nu,\bar{\nu}} \frac{T_i^{D_i}\, e^{-T_i}}{D_i!}. \tag{13}$$

Step 2: log-likelihood ratio. A hypothesis is compared with the best possible description of the data, $T_i = D_i$, through $-2\ln$ of the likelihood ratio [19]:

$$\chi^2(T) = -2\ln\frac{\mathcal{L}(T)}{\mathcal{L}(D)} = 2\sum_i \left[T_i - D_i + D_i \ln\frac{D_i}{T_i}\right]. \tag{14}$$

Step 3: Gaussian limit. Expanding Eq. (14) to second order in $(T_i - D_i)/D_i$ gives

$$\chi^2(T) \simeq \sum_i \frac{(D_i - T_i)^2}{\sigma_i^2}, \qquad \sigma_i^2 = D_i, \tag{15}$$

i.e., each sample has a statistical variance equal to its number of events. The approximation is good for $D_i \gtrsim 10$; for smaller samples Eq. (14) should be used.

Step 4: systematic uncertainty (pull method). A normalization uncertainty f on each sample is introduced as a nuisance parameter a_i, which rescales the prediction, $T_i \to (1+a_i)T_i$, and has a Gaussian prior of width f [23]:

$$\chi^2(T) = \sum_i \min_{a_i} \left[\frac{\big(D_i - (1+a_i)T_i\big)^2}{D_i} + \frac{a_i^2}{f^2} \right] = \sum_i \frac{(D_i - T_i)^2}{D_i + f^2 T_i^2} \simeq \sum_i \frac{(D_i - T_i)^2}{D_i + f^2 D_i^2}, \tag{16}$$

where the minimum over a_i is found analytically and the last step uses $T_i \simeq D_i$. The systematic uncertainty adds in quadrature to the statistical one: it is negligible for $D_i \ll 1/f^2$ and dominant for $D_i \gg 1/f^2$ (e.g., Hyper-K and DUNE with $f = 5\,\%$, $1/f^2 = 400$).

Step 5: expected sensitivity.

- *Data.* For the expected (median) sensitivity the data are replaced by the prediction for the true value, without fluctuations, $D_i = N_i(\delta_{CP})$: the Asimov data set [22]. The best fit is then the true value itself, with $\chi^2 = 0$.
- *Hypothesis to exclude.* CP conservation is the pair of points $\delta' = 0$ and $\delta' = \pi$, with predictions $T_i = N_i(\delta')$. Both must be rejected, so the relevant χ^2 is that of the one that fits best, i.e., the minimum over the two. If the mass ordering is not known, the minimum also includes the CP-conserving points of the other ordering.
- *Significance.* By Wilks' theorem [24] the difference $\Delta\chi^2$ between the χ^2 of the hypothesis and that of the best fit follows a χ^2 distribution with one degree of freedom, and the significance is $n_\sigma = \sqrt{\Delta\chi^2}$: $\Delta\chi^2 = 9$ (25) corresponds to 3σ (5σ) [19].

Final formula. Inserting $D_i = N_i(\delta_{CP})$ and $T_i = N_i(\delta')$ and taking the minimum over the CP-conserving points gives two versions.

(a) Statistics only, no normalization nuisance. From the Gaussian limit, Eq. (15):

$$\boxed{\Delta\chi^2_{\rm stat}(\delta_{CP}) = \min_{\delta' \in \{0,\pi\}} \sum_{i=\nu,\bar\nu} \frac{\big[N_i(\delta_{CP}) - N_i(\delta')\big]^2}{N_i(\delta_{CP})}, \qquad n_\sigma = \sqrt{\Delta\chi^2_{\rm stat}}} \tag{17}$$

This is the sensitivity of a perfectly known normalization (flux, cross section and efficiency without uncertainty): it is limited only by the number of events and grows as $\sqrt{N}$, i.e., as the square root of the exposure.

(b) With a normalization nuisance. From the pull method, Eq. (16):

$$\boxed{\Delta\chi^2(\delta_{CP}) = \min_{\delta' \in \{0,\pi\}} \sum_{i=\nu,\bar\nu} \frac{\big[N_i(\delta_{CP}) - N_i(\delta')\big]^2}{N_i(\delta_{CP}) + \big(f\,N_i(\delta_{CP})\big)^2}, \qquad n_\sigma = \sqrt{\Delta\chi^2}} \tag{18}$$

with N_i from Eq. (5). Eq. (18) reduces to Eq. (17) for $f = 0$. For $N_i \gg 1/f^2$ the denominator is dominated by $(fN_i)^2$ and $\Delta\chi^2$ no longer grows with the exposure: the measurement becomes systematics-limited. In this note $f = 5\,\%$ is used as an illustrative value.

Relation to the experiments' definition. The CP-violation sensitivity is defined, as in the DUNE [26] and Hyper-K [8] design studies, as $\sqrt{\Delta\chi^2}$ with respect to $\delta_{CP} = 0, \pi$ computed on Asimov data. Our version is a simplified counting version: Gaussian statistics, two samples, no backgrounds, fixed oscillation parameters.

Limitations. All other oscillation parameters are fixed at their true values (in a full analysis they are marginalized, which reduces the sensitivity); only two numbers per experiment are used, without the energy spectrum; backgrounds are neglected; Wilks' theorem is only approximately valid for the periodic parameter δ_{CP}, and the experiments use the Feldman–Cousins method [25] for their final results.

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